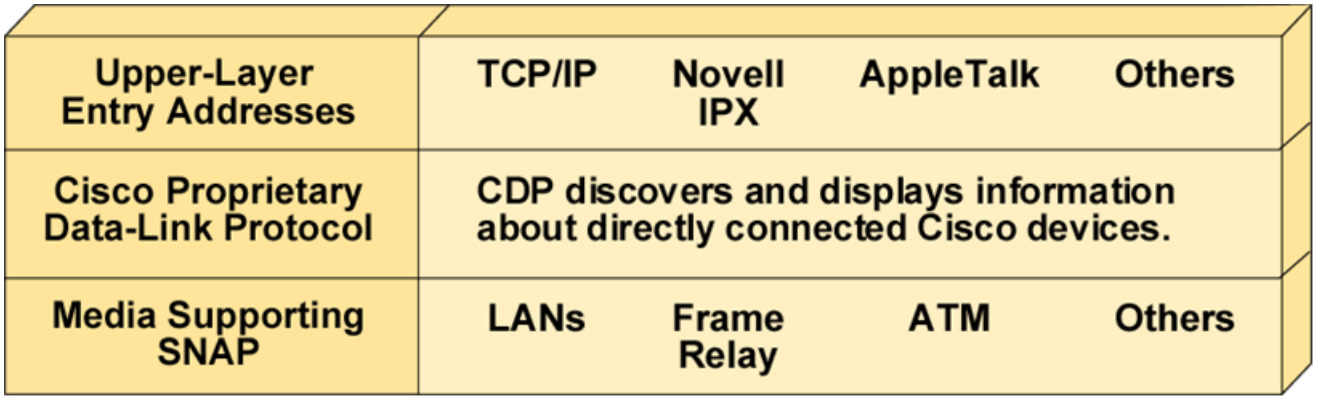
Remote Access Management

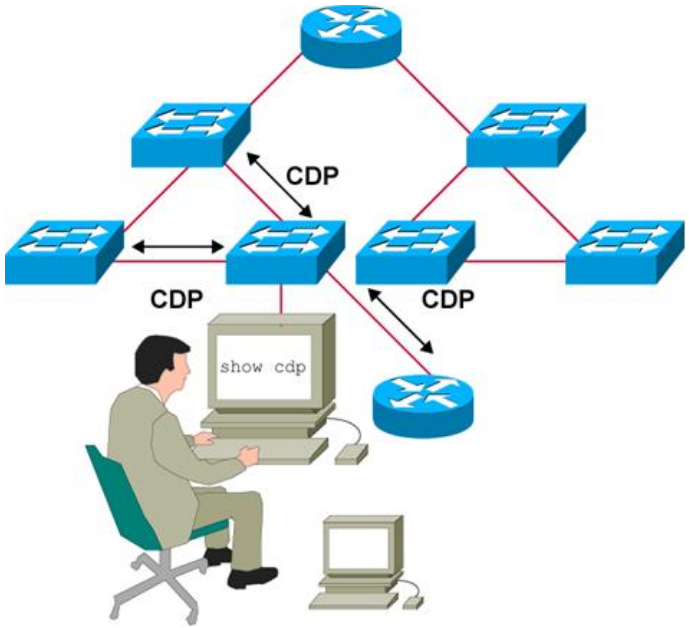
Cisco Discovery Protocol



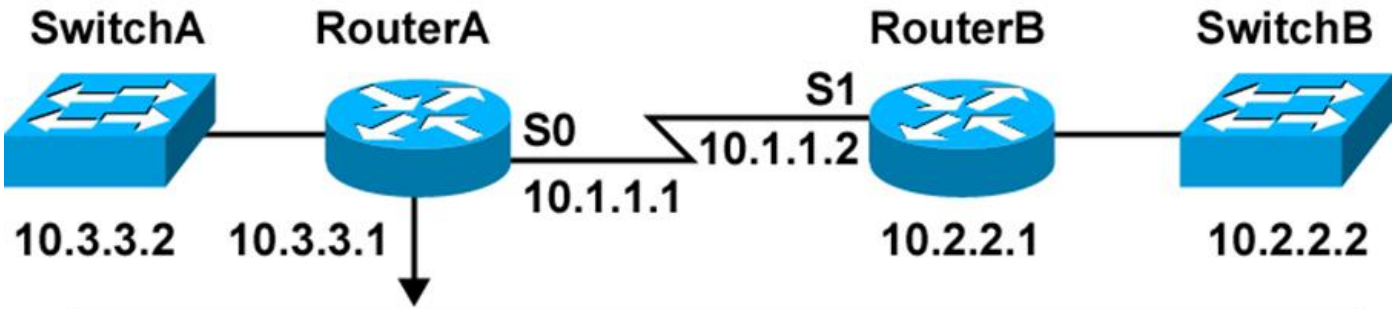
- CDP is a proprietary utility that gives you a summary of directly connected switches, routers, and other Cisco devices.
- CDP discovers neighboring devices regardless of which protocol suite they are running.

Discovering Neighbors with CDP

- CDP runs on routers with Cisco IOS® software Release 10.3 or later and on Cisco switches and hubs.
- Summary information includes:
  - ✓ Device identifiers
  - ✓ Address list
  - ✓ Port identifier
  - ✓ Capabilities list
  - ✓ Platform

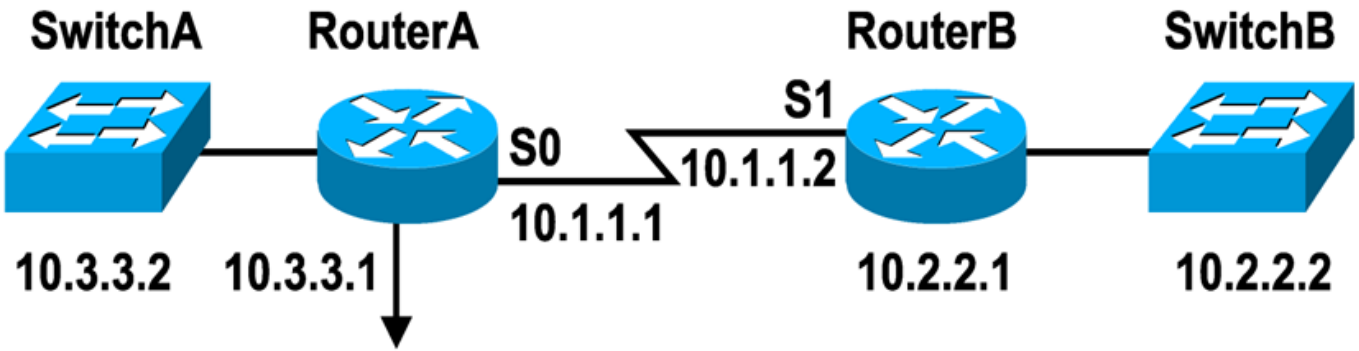


Using CDP




```
RouterA#show cdp ?
  entry      Information for specific neighbor entry
  interface  CDP interface status and configuration
  neighbors  CDP neighbors entries
  traffic    CDP statistics
  <cr>
RouterA(config-if)#exit
RouterA(config)#no cdp run
RouterA(config)#interface serial0
RouterA(config-if)#no cdp enable
```

Using the show cdp neighbors Command

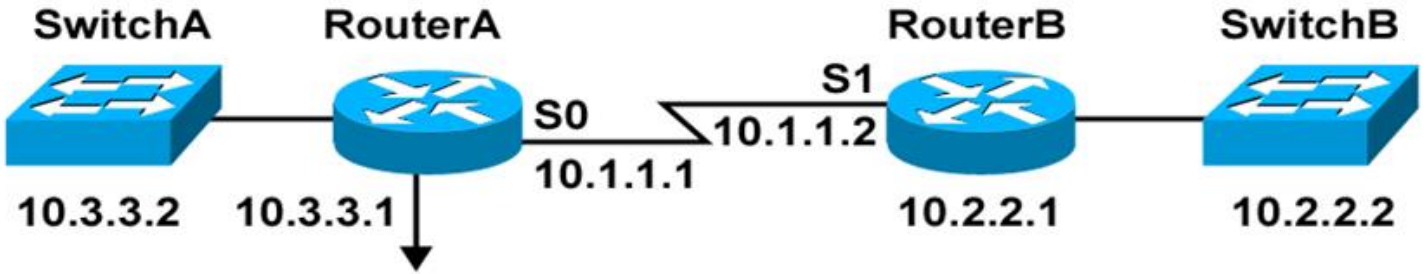



```
RouterA#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
RouterB              Ser 0         148           R        2522      Ser 1
SwitchA0050BD855780  Eth 0         167         T S        1900      2
```

SwitchA also provides its MAC address (Catalyst 1900 only).

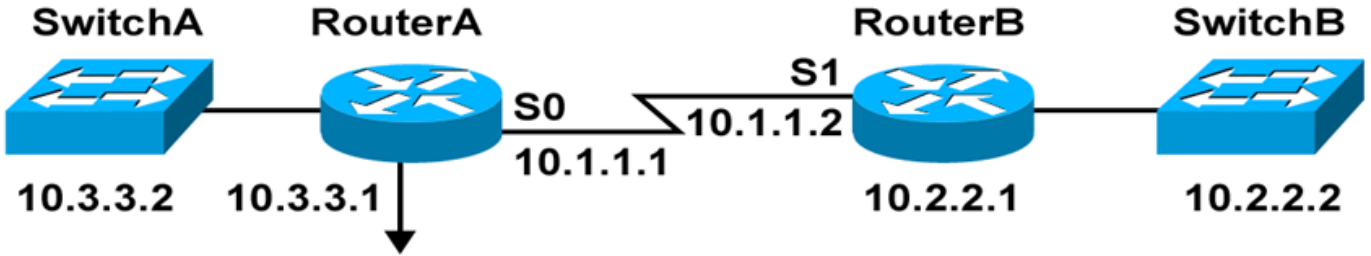
Using the show cdp entry Command




```
RouterA#show cdp entry *
-------------------------
Device ID: RouterB
Entry address(es):
  IP address: 10.1.1.2
Platform: cisco 2522,  Capabilities: Router
Interface: Serial0,  Port ID (outgoing port): Serial1
Holdtime : 168 sec

Version :
Cisco Internetwork Operating System Software
IOS (tm) 2500 Software (C2500-JS-L), Version 12.0(3), RELEASE SOFTWARE (fci)
Copyright (c) 1986-1999 by cisco Systems, Inc.
Compiled Mon 08-Feb-99 18:18 by phanguye
```

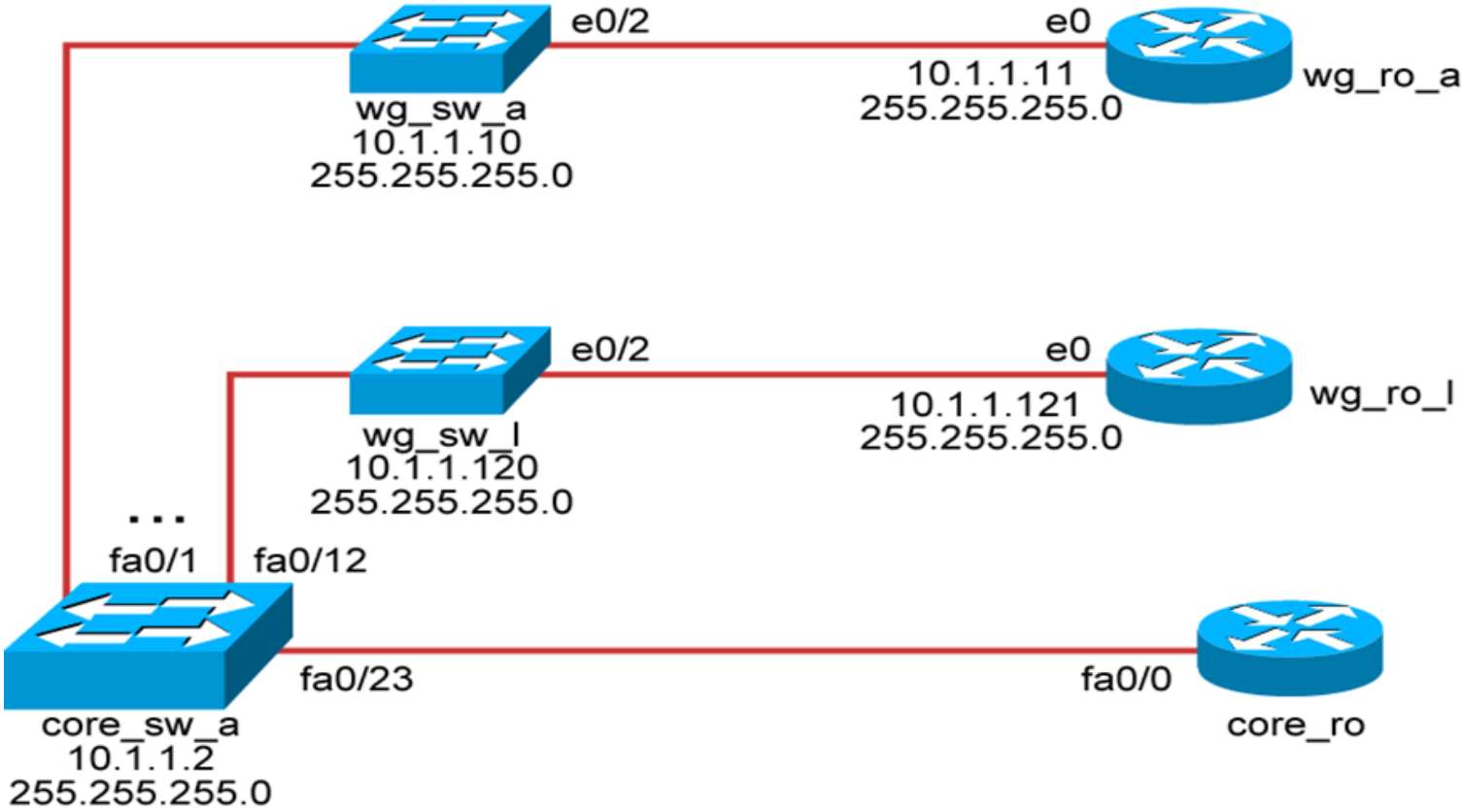
Additional CDP Commands



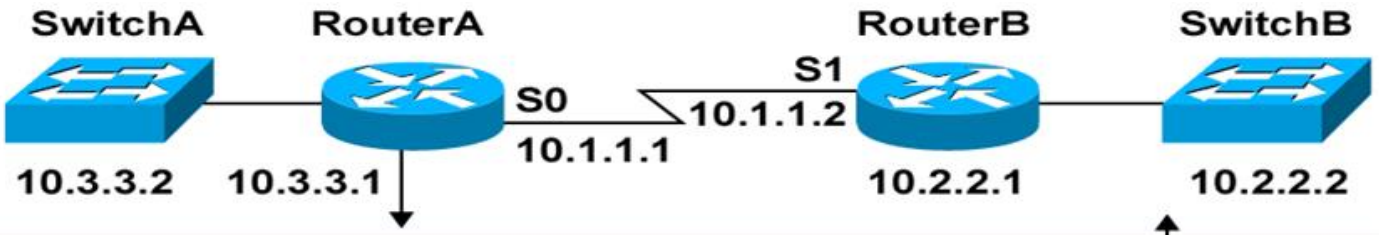

```
RouterA#show cdp traffic
CDP counters :
 Packets output: 56, Input: 38
 Hdr syntax: 0, Chksum error: 0, Encaps failed: 3
 No memory: 0, Invalid packet: 0, Fragmented: 0


RouterA#show cdp interface
BRI0 is administratively down, line protocol is down
  Encapsulation HDLC
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
```

Creating a Network Map



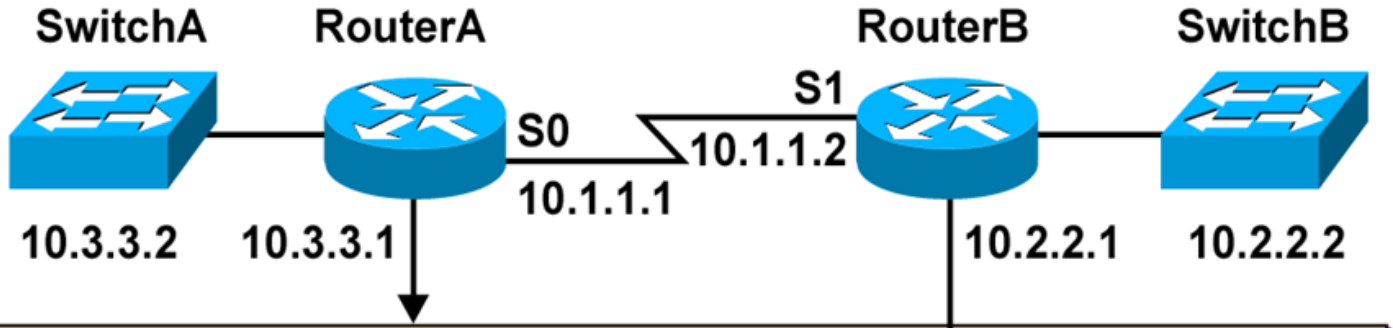
Using Telnet to Connect to Remote Devices




```
RouterA#telnet 10.2.2.2
Trying 10.2.2.2 ... Open
------------------------------------------------
Catalyst 1900 Management Console
Copyright (c) Cisco Systems, Inc.  1993-1998
All rights reserved.
Enterprise Edition Software
Ethernet Address:         00-90-86-73-33-40
PCA Number:               73-2239-06
PCA Serial Number:        FAA02359H8K
Model Number:             WS-C1924-EN
System Serial Number:     FAA0237X0FQ
.
.
SwitchB>
```

Remote device

Viewing Telnet Connections




```
RouterA#show sessions
Conn  Host            Address         Byte    Idle  Conn Name
*  1  10.1.1.2        10.1.1.2        0       0     10.1.1.2
```

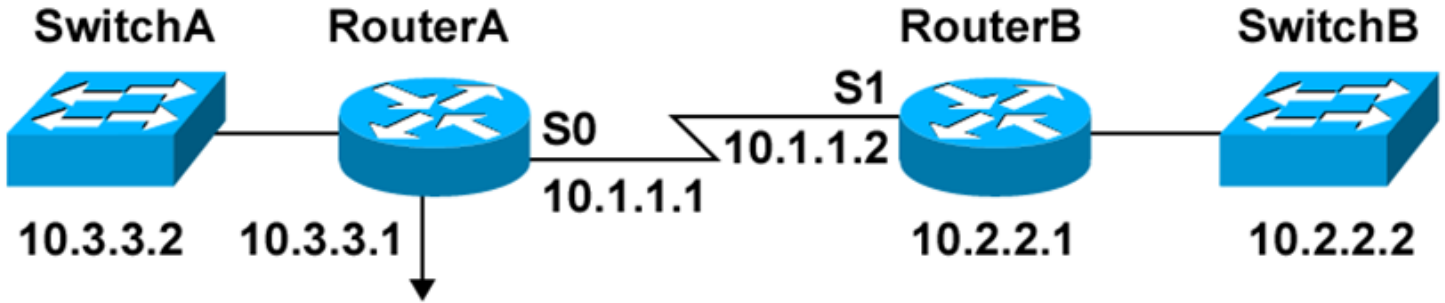
```
RouterB#show users
     Line         User      Host(s)          Idle       Location
*   0 con 0                 idle             1w0d
    11 vty 0                idle           00:00:09     10.1.1.1
```

Suspending and Resuming a Telnet Session




```
RouterB#<Ctrl-Shift-6>x
RouterA#show sessions
Conn Host                Address             Byte  Idle Conn Name
*  1 10.1.1.2            10.1.1.2               0     1 10.1.1.2
RouterA#resume 1
RouterB#
```

Closing a Telnet Session




```
RouterA#disconnect
Closing connection to 10.3.3.2 [confirm]
```

Closes the current session opened by you

```
RouterA#clear line 11
[confirm]
 [OK]
```

Closes a session opened by a remote device

Using the ping and traceroute Commands

```
Router#   ping 10.1.1.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/4/4 ms

Router#   traceroute 192.168.101.101
Type escape sequence to abort.
Tracing the route to 192.168.101.101
  1 p1r1 (192.168.1.49) 20 msec 16 msec 16 msec
  2 p1r2 (192.168.1.18) 48 msec *  44 msec

Router#
```



Telnet vs. SSH Access

- Telnet
  - ✓ Most common access method
  - ✓ Insecure
- SSH - Encrypted

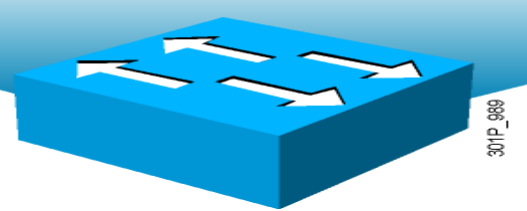
```
!- The username command create the username and password for the SSH session
Username cisco password cisco

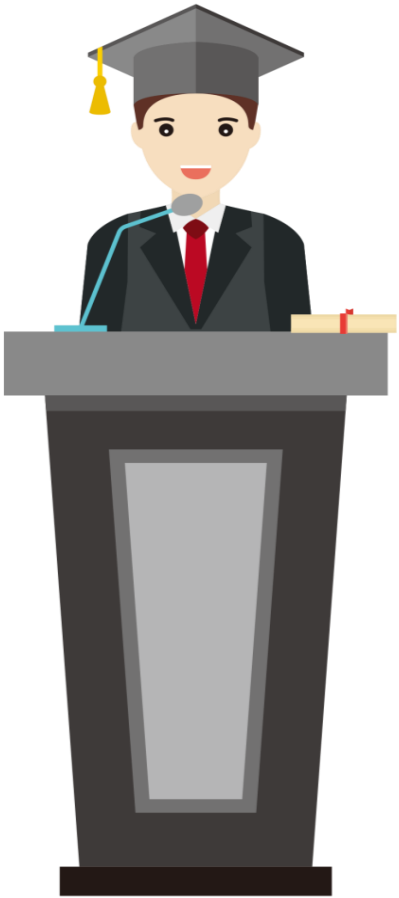
ip domain-name mydomain.com

crypto key generate rsa

ip ssh version 2

line vty 0 4
   login local
   transport input ssh
```





THANK YOU!

